B.Tech CSE - Third Year | Lab 11

Assignment 11: Method Overriding • *Aim: Override inherited methods and demonstrate runtime polymorphism.*

Experiment No. 11.1: Override area() in Square

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Shape {
    int area() { return 0; }
}
class Square extends Shape {
    int side;
    @Override int area() { return side * side; }
}
public class Task_01_Override_area_in_Square {
    public static void main(String[] args) {
        Square s = new Square();
        s.side = 4;
        System.out.println("Area: " + s.area());
    }
}
```

3. **Output:** `Area: 16`
4. **Viva-Voce:** What is method overriding? Why use @Override? What is the area formula for a square?

Experiment No. 11.2: Override area() in Rectangle

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Shape {
    int area() { return 0; }
}
class Rectangle extends Shape {
    int length, breadth;
    @Override int area() { return length * breadth; }
}
public class Task_02_Override_area_in_Rectangle {
    public static void main(String[] args) {
        Rectangle r = new Rectangle();
        r.length = 4; r.breadth = 5;
        System.out.println("Area: " + r.area());
    }
}
```

3. **Output:** `Area: 20`
4. **Viva-Voce:** What is the area formula for a rectangle? How does overriding differ from overloading? What is a superclass?

Experiment No. 11.3: Use polymorphism with Shape reference

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Shape {
    int area() { return 0; }
}
class Square extends Shape {
    int side;
    Square(int side) { this.side = side; }
    @Override int area() { return side * side; }
}
public class Task_03_Use_polymorphism_with_Shape_reference {
    public static void main(String[] args) {
        Shape s = new Square(3);
        System.out.println("Area: " + s.area());
    }
}
```

3. **Output:** `Area: 9`
4. **Viva-Voce:** What is runtime polymorphism? Can a parent reference refer to a child object? Which area() implementation is called?